

```
In [7]: pip install numpy
```

Requirement already satisfied: numpy in .\anaconda3\Lib\site-packages (2.4.6)

Note: you may need to restart the kernel to use updated packages.

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt

import numpy as np
```

```
In [2]: dataset =pd.read_csv("50_Startups_dataset.csv")
```

```
In [3]: dataset =pd.get_dummies(dataset,drop_first=True)
```

```
In [4]: independent = dataset[['R&D Spend', 'Administration', 'Marketing Spend'],'
dependent = dataset[['Profit']]
```

```
In [96]: from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(independent,dependent,test
```

```
In [97]: from sklearn.preprocessing import StandardScaler
```

```
sc = StandardScaler()

x_train = sc.fit_transform(x_train)
x_test = sc.transform(x_test)
```

```
In [98]: from sklearn.svm import SVR
regressor=SVR(kernel="linear",C=10000)
regressor=regressor.fit(x_train,y_train)
```

C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

```
y = column_or_1d(y, warn=True)
```

```
In [99]: y_pred=regressor.predict(x_test)
y_pred
```

```
Out[99]: array([117370.67464976,  91227.63153915,  78739.73678098,  70838.6758226
7,
               176434.87701777, 171646.5117204 ,  48852.81324443, 103993.1088376
9,
               60235.80513728,  99963.84485193])
```

```
In [100]: from sklearn.metrics import r2_score
percentage=r2_score(y_test,y_pred)*100
print(f"result: {percentage} %")
```

result: 95.88705258814116 %

```
In [101]: import pickle
filename = 'PaL_model.sav'
pickle.dump(regressor, open(filename, 'wb'))
```

```
In [102... loaded_model = pickle.load(open('P&L_model.sav', 'rb'))
result = loaded_model.predict([[1000,59999,6969,0,1]])
print(result)
```

```
[[47053.13025183]]
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\validation.
py:2827: UserWarning: X does not have valid feature names, but LinearRegre
ssion was fitted with feature names
  warnings.warn(
```

```
In [103... import pickle

loaded_model = pickle.load(open('P&L_model.sav', 'rb'))

rd_spend = float(input("Enter R&D Spend: "))
administration = float(input("Enter Administration cost: "))
marketing_spend = float(input("Enter Marketing Spend: "))

state = input("Enter State (California/Florida/New York): ").lower()

if state == "florida":
    state_florida = 1
    state_new_york = 0
    state_name = "Florida"

elif state == "new york":
    state_florida = 0
    state_new_york = 1
    state_name = "New York"

elif state == "california":
    state_florida = 0
    state_new_york = 0
    state_name = "California"

else:
    print("Invalid State")
    exit()

prediction_input = [[
    rd_spend,
    administration,
    marketing_spend,
    state_florida,
    state_new_york
]]

result = loaded_model.predict(prediction_input)
print(f"For {state_name} predicted profit is {result[0]}")

if predicted_input < 0:
    print(f"For {state_name}, the predicted result is a loss of {(predict
elif predicted_profit > 0:
    print(f"For {state_name}, the predicted profit is {predicted_input}")
else:
    print(f"For {state_name}, there is no profit or loss.")
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
For New York predicted profit is [52645.99020431]
```

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-
NameError                                Traceback (most recent call last)
Cell In[103], line 42
     38
     39 result = loaded_model.predict(prediction_input)
     40 print(f"For {state_name} predicted profit is {result[0]}")
     41
--> 42 if predicted_input < 0:
     43     print(f"For {state_name}, the predicted result is a loss of {(predicted_input)}")
     44 elif predicted_profit > 0:
     45     print(f"For {state_name}, the predicted profit is {predicted_input}")

NameError: name 'predicted_input' is not defined
```

In []: